

Option-Only Markowitz as Conditional Option Cashflow Engineering

Dhruv Kohli

Abstract

How should an allocator choose among listed calls and puts when the instruments are funded, expiring, state-contingent cashflows rather than ordinary asset return columns? This paper builds the option-only analogue of Markowitz for that problem. Each option is represented by premium-funded NAV exposure, payoff and settlement convention, conditional option-risk-premium forecast, Greek exposure, implementation screen, and residual risk. The covariance matrix is induced by the Greek map,

$$r_O = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \quad \Sigma_{O,t} = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t},$$

and the optimizer applies the familiar efficient-frontier algebra to premium weights scaled by portfolio NAV.

In this sample and under these conventions, the empirical evidence is intentionally mixed. The framework is coherent, point-in-time, and auditable. Gross option-only portfolios can look attractive, but gross Sharpe is a research diagnostic, not a trading result. VIX options can improve the design as volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the relevant rows. Post-cost accounting, liquidity screens, beta and Greek budgets, inference, and short-option risk controls make the deployable interpretation much more conditional. The portfolio-construction contribution is therefore not a live-trading claim. It is a disciplined way to decide whether convexity, volatility exposure, or short-premium risk deserves capital after premium accounting, costs, settlement, beta audits, Greek budgets, and out-of-sample validation.

Key Findings

- Option-only Markowitz is feasible when options are treated as premium-funded cashflows with Greek-induced covariance, not as generic return columns.
- Gross option portfolios can appear attractive, but post-cost research accounting, settlement gates, and short-option risk controls sharply limit what can be claimed.
- VIX options are useful as conditional volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the evidence.

Impact Statement

Option research often begins with a named trade: covered calls, crash puts, variance spreads, or volatility hedges. This paper begins one layer earlier. It gives a reproducible template for turning arbitrary listed calls and puts into NAV-normalized cashflows, Greek exposures, conditional premia, covariance estimates, and constraints before asking whether they deserve capital. The conclusion is useful precisely because it is restrained: option-only Markowitz is mathematically clean, but empirical claims survive only if premium accounting, settlement, costs, inference, and equity-beta audits support them.

1. Introduction

An investor restricted to listed options has a simple question with a difficult answer: which calls and puts actually deserve premium capital? The usual asset-allocation template starts from return columns. Options do not fit that template cleanly. They are funded, expiring, state-contingent cashflows whose economics depend on premium paid or received, strike, expiry, exercise style, settlement, volatility, convexity, and the state of the underlying.

This paper builds a premium-weighted option-only Markowitz framework for that choice. The universe can contain arbitrary listed calls and puts across equities, strikes, expiries, and moneyness buckets. Positions can be long or short. Cash is not optimized as a separate risky asset; it is the NAV numeraire and collateral account used to scale option premium exposures. The premium is still paid or received. A long option that expires worthless loses its premium weight, and a short option that finishes in the money pays the payoff through the same NAV accounting. The allocation problem is therefore not “which stock return column has the best mean,” but whether a funded option cashflow deserves capital after settlement, costs, beta exposure, Greek budgets, and validation gates.

1.1. Research Question and Contributions

The research question is: how should an allocator choose among listed calls and puts when every instrument is a funded, expiring, state-contingent cashflow rather than a standard asset return column? The proposed answer is a premium-weighted option-only Markowitz framework that maps listed options into NAV-normalized cashflows, payoff and settlement conventions, Greeks, residual risk, conditional option-risk-premium forecasts, implementation screens, and portfolio constraints. In this sample and under these conventions, the validated empirical claim is that the framework is coherent, point-in-time, auditable, and useful for diagnosing whether option convexity, VIX exposure, or short-premium risk deserves capital. The non-claim is equally important: the article does not claim live alpha, production tradability, broker-executed performance, live margin parity, or deployable option trading performance.

The organizing thesis is that option-only portfolio construction should be written as conditional option cashflow engineering. A listed option is not a stock with a different ticker. It is a state-contingent claim with delta, gamma, vega, theta, exercise, settlement, liquidity, and marking conventions. Two options can share the same underlying and still contribute very different risks. A put and a call can leave the book with little net delta and large volatility or

convexity exposure. The covariance matrix must therefore come from the instrument map, not only from historical option-price correlations.

The theoretical contribution is the option-only analogue of Markowitz. Classical portfolio theory starts with expected returns and an asset covariance matrix (Markowitz, 1952). This paper starts with option premium weights, a local Greek return map, conditional option-risk-premium forecasts, and a residual covariance term. Once those objects are built, the mean–variance algebra is familiar: the unconstrained maximum–Sharpe direction is proportional to $\hat{\Sigma}_{O,t}^{-1} \hat{\mu}_t$. The difference is the construction of $\hat{\Sigma}_{O,t}$ and $\hat{\mu}_t$, not the final quadratic geometry.

What is new is the object rather than a single trade. The paper is not a defense of long calls, short puts, covered calls, protective puts, or a named volatility spread. It asks how to allocate across an arbitrary option universe once each contract has a mark, a payoff convention, a Greek exposure vector, a conditional expected return, and an implementation ledger. The expected-return side is also explicit: covariance optimization alone cannot create a high Sharpe ratio. A useful option allocator needs forecastable premia such as carry, variance risk premia, volatility risk premia, skew compensation, tail-insurance pricing, relative value, and regime-dependent convexity.

1.2. Related Work and Positioning

The starting point is mean–variance allocation. Markowitz (1952) showed how expected returns and a covariance matrix define an efficient frontier, while Sharpe (1966) made reward-to-variability central to performance evaluation. Listed options require a different first step because the columns are not ordinary asset returns. Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973) and the Black–76 forward option model (Black, 1976) provide local pricing and Greek maps, not a complete physical-measure portfolio allocation rule. The present framework uses those maps to translate options into delta, gamma, vega, theta, VIX-forward exposure, and residual risk before applying Markowitz geometry.

Expected option returns have their own economic sources. Empirical option-return research links compensation to carry, volatility risk premia, skew, crash insurance, and tail-risk states (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). This article uses that spine to motivate conditional option-risk-premium forecasts rather than treating the optimizer as a source of alpha. Robust covariance estimation also matters because inverse-covariance portfolio rules amplify noisy inputs; shrinkage follows Ledoit and Wolf (2004). Evaluation uses both reward-to-volatility and downside-aware measures, including Sortino (Sortino and Price, 1994), Omega (Keating and Shadwick, 2002), Calmar and drawdown (Magdon-Ismail and Atiya, 2004), and active risk (Grinold and Kahn, 2000).

Implementation frictions and validation discipline determine the strength of any empirical claim. Option overlays are sensitive to spread, settlement, borrow, assignment, margin, and capacity assumptions. Backtest overfitting and multiple testing can turn optimizer variants into false discoveries, so the analysis uses bootstrap inference and reality-check ideas consistent with Bailey et al. (2017) and López de Prado (2018). The distinction from strategy-specific option papers is therefore deliberate. The object is not a covered-call, put-write, crash-put, or volatility-spread test. The object is an instrument-level cashflow accounting and allocation framework for arbitrary listed options before asking whether any option sleeve

deserves capital.

The empirical result is intentionally mixed. The gross point-in-time backtest tests whether the option-only Markowitz object behaves coherently on the local OPRA-derived panel. The post-cost research layer subtracts conservative spreads, fees, slippage, borrow, capacity, simulated margin funding, and assignment-risk penalties. VIX options are included as volatility derivatives with VX-forward Black-76 Greeks, but VIX option evidence is headline-grade only when exact VRO/SOQ settlement supports the relevant rows. If the coverage table reports proxy rows, the same evidence is reported only as diagnostic hedge-sleeve evidence rather than a listed-settlement claim.

The article proceeds in the same order as the research workflow. Section 2 defines the data and empirical design. Section 3 builds the option cashflow and Greek-risk framework. Section 4 describes the conditional portfolio construction model. Section 5 sets out the metrics, point-in-time validation, inference, and claim gates. Section 6 reports the empirical results and negative evidence. Section 7 concludes with the limits of the current research implementation. Appendix A contains supplemental figures, option/VIX diagnostics, robustness tables, instrument conventions, proof sketches, and reproducibility notes.

2. Data

The empirical design is a monthly, point-in-time allocation study over listed equity options and VIX options. The investable equity-option universe is restricted to AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA, using OPRA/Databento-derived option panels after basic mark, expiry, Greek, and liquidity filters. The VIX-option universe is built from raw VIX chain shards and kept separate because VIX options are volatility derivatives, not options on an equity spot price. Table 1 records the sample size, observation count, train/test split, settlement status, and post-cost layer used in the baseline run.

Item	Value
Raw OPRA-derived equity option rows	160,315
Raw VIX option rows after filters	103,650
Monthly snapshot dates	129
Training dates	69
Test dates	60
Primary equity underlyings	AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, TSLA
VIX option treatment	VX-forward Black-76 Greeks; VIX/VVIX state only
VIX settlement source	vro_soq_exact:536
VIX headline status	headline-grade exact VRO/SOQ
Post-cost layer	pre-production research simulation: mid, half-spread, and full-spread executable-cost scenarios
Broker execution status	not broker-executed live evidence
Rolling option bucket assets	54
Train/test split	through 2020-12-31 / after 2020-12-31

Table 1: Empirical design summary. The table identifies the option universe, monthly snapshots, train/test split, VIX settlement status, and post-cost convention used in this sample and under these conventions.

The train/test split is frozen at December 31, 2020. Training observations through that date estimate expected option returns, covariance objects, residual risk, factor exposures, and implementation screens. Test observations after that date are reserved for out-of-sample portfolio evaluation. Rebalancing is monthly. At each decision date, the research implementation sees only the option panel, state variables, underlying prices, VX curve, forecast inputs, and cost inputs observable at or before that date. The realized option return is then measured at

the next monthly realization date or at listed expiry, according to the holding ledger.

Equity-option rolling buckets convert expiring contracts into stable empirical objects without pretending that options are perpetual assets. For each underlying, right, moneyness, tenor, and liquidity bucket, the decision date selects a representative listed contract using contemporaneous marks and filters. The selected option can be a call or a put, and the optimizer can hold it long or short subject to gross NAV, concentration, Greek, beta, short-premium, liquidity, stress, and assignment-risk constraints. The return includes the premium paid or received: a long option that expires worthless loses its premium weight, while a short option that finishes in the money pays the realized payoff through NAV accounting.

The VIX-option construction follows a different convention. Raw VIX chain shards are stacked, OSI symbols are parsed, duplicate (trade date, symbol) rows are coalesced, and each option is aligned to the relevant VX futures forward. Black-76 Greeks are computed with respect to the VX forward rather than spot VIX. VIX, VVIX, VIX9D, and VIX3M enter only as state variables. Exact VRO/SOQ settlement is a claim gate: a VIX option becomes headline-grade only when the expiry row uses `vro_soq_exact`. Proxy rows, if any, remain appendix diagnostics and cannot support listed-settlement claims.

Settlement source	Rows	Headline eligible
<code>vro_soq_exact</code>	536	yes

Table 2: VIX settlement coverage. VIX option evidence is headline-grade only when every VIX expiry row used in headline tables has `vro_soq_exact` settlement.

Marks, payoffs, settlement, and costs are treated as first-order design choices. Option weights are premium exposures per dollar of NAV. The mark source, payoff date, settlement source, Greek model, state snapshot, and train-end date are recorded before performance is reported. Post-cost research returns use three executable-cost scenarios, explicit fees, spread haircuts, no-fill rules, capacity limits, simulated margin/stress capital, borrow proxies, and assignment-risk flags. These conventions are adequate for a research paper on allocation mechanics because they test whether the option-only Markowitz object can be constructed, audited, stress-screened, and evaluated out of sample. They are not adequate for production trading claims because they do not contain broker-routed orders, live quote depth, fill probabilities calibrated from orders, live margin previews, or broker reconciliation.

Object	Accounting rule	Main failure if omitted
Equity option	Premium weight times mark or payoff return	Treats the option as free convexity
Short option	Premium received plus future liability	Hides tail and assignment exposure
VIX option	Black-76 exposure to matched VX forward	Treats spot VIX as tradable
Cash/collateral	NAV numeraire, not optimized asset	Converts the model into a stock-cash allocation
Post-cost layer	Spreads, fees, slippage, borrow, capacity, margin, assignment flags	Promotes gross research returns to false trading evidence

Table 3: Option accounting contract. Each object is represented by the cashflow convention that must be specified before the optimizer can compare it with another option.

3. Option Cashflow and Risk Framework

The central modeling step is to express every option as a NAV-normalized cashflow. Let W_t be beginning portfolio NAV, $q_{i,t}$ the signed premium weight assigned to option i , and $C_{i,t} > 0$ the decision-date mark. The contract count is

$$N_{i,t} = \frac{q_{i,t} W_t}{C_{i,t}}.$$

Ignoring transaction costs and slippage for the gross theory object, but not ignoring the option purchase price, the next-period NAV contribution is

$$\frac{\Delta W_{i,t+1}}{W_t} = q_{i,t} \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}}.$$

These are premium weights. A long option whose mark falls to zero loses exactly the paid premium weight. A short option is scaled by the premium it sold and then pays the future mark or payoff liability.

For a call or put on equity underlying u , the local mark change is approximated by the same Ito–Taylor accounting that underlies Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973; Shreve, 2004):

$$\Delta C_i \approx \Delta_i \Delta S_u + \frac{1}{2} \Gamma_i (\Delta S_u)^2 + \nu_i \Delta \sigma_i + \Theta_i \Delta t + \eta_i.$$

The paper does not require Black–Scholes to be a globally correct pricing model for American single-name options. It uses the weaker local statement: an option mark can be expanded in spot, volatility, and time, with residual marking error left in η_i .

Dividing by the option mark gives the return exposure map

$$r_i = \underbrace{\frac{\Delta_i S_u}{C_i}}_{b_{i,u}^\Delta} R_u + \underbrace{\frac{1}{2} \frac{\Gamma_i S_u^2}{C_i}}_{b_{i,u}^\Gamma} R_u^2 + \underbrace{\frac{\nu_i}{C_i} \Delta \sigma_i}_{b_{i,u}^\nu} + \underbrace{\frac{\Theta_i \Delta t}{C_i}}_{\theta_i^*} + \epsilon_i.$$

The coefficients are dimensionless option-return exposures. Deep out-of-the-money options can have large return exposure because the premium denominator is small. That is not a numerical nuisance. It is the reason option-only portfolios need explicit gross, per-contract, Greek, beta, and stress budgets.

Stacking contracts gives the factor form

$$r_O = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \quad \Sigma_{O,t} = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t}.$$

Here f contains underlying returns, centered squared returns, and implied-volatility shocks; B_t contains delta-return, gamma-return, and vega-return loadings; and $\Sigma_{\varepsilon,t}$ is the residual option-return covariance. The residual term is part of the model. Omitting it would assume that Greeks span all marking risk.

VIX options require a separate convention. They are not options on an equity spot. The

traded forward-risk anchor is the matched VX futures curve, and Black's futures-option logic is the natural local model (Black, 1976). For VIX option j , the local approximation is

$$\Delta C_j^{VIX} \approx \Delta_j^{VX} \Delta F_{T_j}^{VX} + \frac{1}{2} \Gamma_j^{VX} (\Delta F_{T_j}^{VX})^2 + \nu_j^{VIX} \Delta \sigma_j^{VIX} + \Theta_j^{VIX} \Delta t.$$

VIX, VVIX, and term-structure indices are useful state variables; VX futures and VIX options are the traded volatility instruments. Exact VRO/SOQ settlement is required before VIX option expiry P&L can be promoted from settlement proxy to headline listed-settlement evidence.

4. Portfolio Construction Model

The optimizer is chosen only after the cashflow and risk map are built. Covariance is not enough. A high Sharpe ratio cannot come from $\Sigma_{O,t}$ alone; it must come from a conditional expected-return vector. The paper writes the option mean as

$$\hat{\mu}_{i,t} = b_{i,t}^S \hat{\lambda}_t^S + b_{i,t}^\Gamma \hat{\lambda}_t^{var} + b_{i,t}^\nu \hat{\lambda}_t^{vol} + b_{i,t}^{skew} \hat{\lambda}_t^{skew} + b_{i,t}^{tail} \hat{\lambda}_t^{tail} + \theta_{i,t}^*.$$

Delta loads on equity compensation. Gamma and vega load on variance and volatility risk premia. Put skew and VIX call wings load on crash-insurance and tail premia. Theta is carry under unchanged state variables. The option-return literature documents why these components can carry expected returns rather than only hedge noise (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). In implementation the combined signal is estimated only from training-window or decision-date information and is shrunk toward zero.

Given $\hat{\mu}_t$ and $\hat{\Sigma}_t$, the gross option-only Sharpe problem is

$$\max_{q \in \mathcal{K}_t} \frac{q^\top \hat{\mu}_t}{\sqrt{q^\top \hat{\Sigma}_t q}}, \quad \hat{\Sigma}_t = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t}.$$

If the feasible set is unconstrained and $\hat{\Sigma}_t$ is positive definite, the maximum-Sharpe direction is proportional to $\hat{\Sigma}_t^{-1} \hat{\mu}_t$. This is exactly the Markowitz tangency formula. The option work is in building the conditional option return object that the formula consumes.

The practical feasible set is smaller. The research implementation imposes gross NAV, short premium, per-contract, underlying concentration, delta, SPY beta, eight-underlying beta, gamma, equity vega, VIX vega, and scenario stress-loss budgets where the corresponding strategy requires them. The optimizer can therefore choose calls, puts, long positions, and short positions, but it cannot hide large exposure behind a small option premium.

Strategy	Predicted monthly vol	Train realized vol	Test realized vol	Test / predicted
Equity-option Greek Markowitz	0.334	0.267	0.307	0.918
Greek Markowitz + VIX	0.363	0.261	0.262	0.722
Beta/delta-neutral + VIX	0.354	0.272	0.239	0.677
Equal premium	1.509	0.625	0.542	0.359
Equal risk	0.668	0.394	0.368	0.551
VIX hedge sleeve	4.648	4.092	0.417	0.090

Table 4: Greek-induced covariance calibration. Predicted monthly volatility is computed from $q^\top \Sigma_O q$ using the training-window Greek risk model; realized volatility is measured on train and test option returns.

The estimator follows the standard risk-model route. It constructs B_t , estimates Ω_t from point-in-time state shocks, estimates $\Sigma_{\varepsilon,t}$ from training residuals, shrinks the covariance toward a diagonal target, and repairs small negative eigenvalues. This follows the same concern that motivates covariance shrinkage in ordinary portfolio problems: noisy matrices become dangerous when the optimizer inverts them (Ledoit and Wolf, 2004).

5. Metrics and Validation

The validation design asks several different questions. The realized path asks what happened in the held-out monthly window. The point-in-time audit asks whether decisions used only information available at the rebalance date. The regression asks whether the option book is mostly equity drift or volatility exposure. The attribution asks whether P&L comes from delta, gamma, vega, carry, VIX-forward exposure, skew/tail terms, or residual marking error. Cost stress asks whether the gross result survives realistic research frictions.

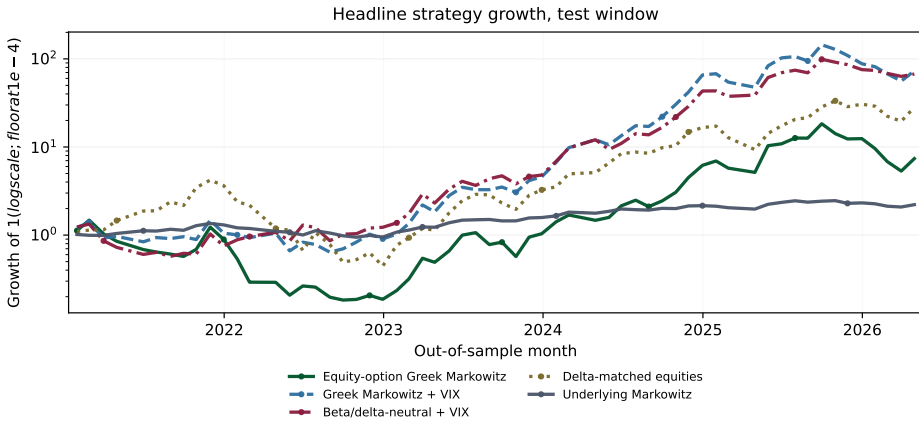


Figure 1: Headline out-of-sample growth paths. The figure focuses on the central option allocators and equity benchmarks; the all-strategy growth diagnostic appears in Appendix A. The log-scale floor prevents failed strategies from disappearing visually.

Performance is reported with complementary measures. Sharpe follows the reward-to-variability tradition (Sharpe, 1966). Sortino replaces total volatility with downside deviation (Sortino and Price, 1994). Calmar scales return by maximum drawdown (Magdon-Ismael and

Atiya, 2004). Omega compares gains above a threshold with shortfalls below it (Keating and Shadwick, 2002). Information ratio measures active return against SPY (Grinold and Kahn, 2000). The zero monthly threshold is used for Sortino and Omega because cash is a numeraire in the model, not an alpha sleeve.

The inference layer is intentionally conservative. Sharpe, Sortino, Calmar, Omega, regime statistics, and leave-one-out differences are reported with block-bootstrap intervals where meaningful. Factor alpha uses HAC/Newey–West standard errors. These intervals do not prove economic significance, but they prevent the paper from reading point estimates as certainties. The pre-production extension also reports probabilistic Sharpe ratios, deflated Sharpe ratios, and a block-bootstrap max-Sharpe screen across the tested gross and executable-cost variants.

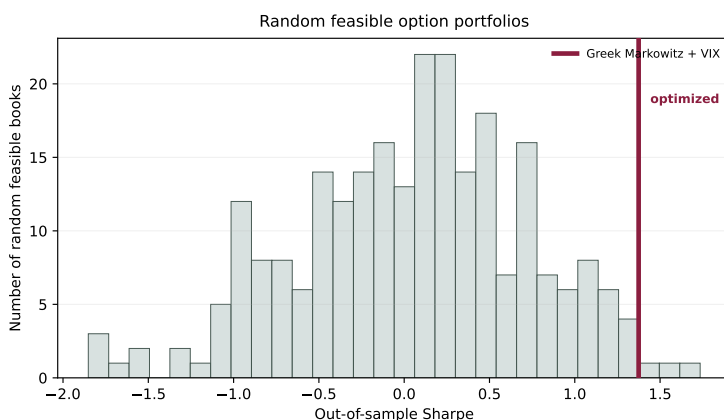


Figure 2: Random feasible comparison. The histogram shows out-of-sample Sharpe ratios for random gross-NAV-feasible option books; the vertical marker shows the optimized Greek-Markowitz-plus-VIX sleeve. The exhibit asks whether optimization adds information beyond arbitrary feasible option exposure.

The point-in-time contract is strict. Decision dates precede return dates. State snapshots, Greeks, forecasts, covariances, residuals, betas, borrow proxies, and cost inputs are drawn from the training window or the decision date. Expiry spot, future IV, future VIX/VVIX, future VX curves, and future option returns are not allowed to enter portfolio construction. Cost-scenario ledgers, required-capital denominators, hurdle/no-trade records, liquidity-tier diagnostics, assignment-risk filters, and reality-check inputs obey the same rule. The audit layer reads the machine-readable outputs independently and fails if core lineage, settlement, cost, inference, claim, or manuscript checks fail.

6. Results and Discussion

The strongest result is conceptual rather than promotional. In this sample and under these conventions, the option-only Markowitz object is coherent, auditable, and measurable. The empirical investment result is more conditional. Gross option portfolios can look attractive, but gross Sharpe is a research diagnostic rather than a trading result. VIX options improve the design as volatility instruments and hedge candidates only when exact VRO/SOQ settlement coverage supports the headline rows.

The body keeps the central evidence in view: out-of-sample performance, full-spread post-cost survival, factor exposure, the realized path in Figure 1, the random feasible comparison in Figure 2, the VIX settlement gate in Table 2, claim strength, and the claim audit. Detailed liquidity, attribution, regime, forecast-ablation, and multiple-testing diagnostics appear in Appendix A.

6.1. Headline evidence

Return basis	Strategy	Ann. return	Ann. vol	Sharpe	Sortino	Calmar	Omega	Info. ratio
Gross before costs	Equity-option Greek Markowitz	0.894	1.062	0.842	1.770	1.020	1.888	0.598
Gross before costs	Greek Markowitz + VIX	1.246	0.907	1.374	3.385	2.034	2.784	1.200
Gross before costs	Beta/delta-neutral + VIX	1.172	0.829	1.414	3.410	2.057	2.948	1.147
Gross before costs	Equal premium	-0.056	1.878	-0.030	-0.055	-0.056	0.977	-0.120
Gross before costs	Equal risk	0.104	1.275	0.082	0.146	0.107	1.064	-0.058
Gross before costs	VIX hedge sleeve	-9.285	1.444	-6.429	-3.089		0.040	-6.132
Gross before costs	Delta-matched equities	1.095	0.931	1.176	2.317	1.225	2.367	1.041
Gross before costs	Underlying Markowitz	0.179	0.196	0.909	1.856	0.590	1.929	0.072
Post-cost research	Equity-option Greek Markowitz	-7.900	5.476	-1.443	-1.347	-6.357	0.098	-1.545
Post-cost research	Greek Markowitz + VIX	-6.079	4.827	-1.259	-1.198	-5.136	0.116	-1.345
Post-cost research	Beta/delta-neutral + VIX	-3.896	2.841	-1.371	-1.300	-3.895	0.151	-1.508
Post-cost research	Equal premium	-3.165	2.378	-1.331	-1.502	-3.051	0.359	-1.372
Post-cost research	Equal risk	-1.276	1.367	-0.934	-1.199	-1.276	0.505	-1.082
Post-cost research	VIX hedge sleeve	-14.364	4.209	-3.413	-2.445	-12.391	0.016	-3.357
Post-cost research	Delta-matched equities	1.095	0.931	1.176	2.317	1.225	2.367	1.041
Post-cost research	Underlying Markowitz	0.179	0.196	0.909	1.856	0.590	1.929	0.072

Table 5: Gross and post-cost out-of-sample performance. Gross rows test the option-only Markowitz object; post-cost rows are research simulations that subtract spreads, fees, slippage, borrow, capacity, simulated margin funding, and assignment-risk penalties.

The table separates two questions. The gross rows ask whether the mathematical allocation object works on held-out option returns. The post-cost rows ask whether the same object still looks economically attractive after conservative implementation frictions. The pre-production layer then asks whether the result survives midpoint, half-spread, and full-spread executable-cost scenarios, required-capital denominators, no-fill diagnostics, and margin/stress-capital screens. Gross Sharpe is not treated as a trading result.

Strategy	Gross Sharpe	Net Sharpe	Net Calmar	Avg spread cost	Capacity used	Survives?
Equity-option Greek Markowitz	0.8421194565895301	0.7668067218645858	0.9148415644409728	0.004454581057950615	0.9901002345178692	yes
Greek Markowitz + VIX	1.3743885779509968	1.0258219183556512	1.331603783957096	0.02189086589088895	0.9988451122012506	yes
Beta/delta-neutral + VIX	1.4135236431740843	0.9508576292459209	1.2239969998479348	0.026468684884072495	0.9997049345720702	yes
Equal premium	-0.02955676336526383	-0.20462197552298175	-0.38425395579926913	0.020340974748224574	0.9993803841618195	no
Equal risk	0.0818308658646486	-0.11815498556113241	-0.15161444631136373	0.015983091604746466	0.9990134907215874	no
VIX hedge sleeve	-6.429184989935395	-6.607601788466617	nan	0.022500000000000003	0.9885078059495911	no

Table 6: Post-cost survival under the full-spread executable-cost scenario. A strategy survives only if net performance remains positive after conservative research frictions, capacity screens, settlement gates, and risk-capital checks.

Strategy	Ann. alpha	Alpha HAC t	R^2	Residual ann. vol	Beta SPY	Beta VX front	Beta dVIX	Beta dVVIX
Equity-option Greek Markowitz	0.405	1.141	0.618	0.634	0.005	-1.368	0.046	0.003
Greek Markowitz + VIX	0.675	2.060	0.513	0.616	1.908	-0.767	0.056	-0.001
Beta/delta-neutral + VIX	0.845	2.753	0.429	0.601	0.715	-0.522	0.038	-0.001
Equal premium	-0.416	-0.566	0.253	1.647	2.522	-0.749	0.032	-0.001
Equal risk	-0.183	-0.334	0.253	1.115	2.041	-0.379	0.020	-0.003
VIX hedge sleeve	-8.807	-14.623	0.252	1.264	3.562	-1.180	0.093	-0.013
Delta-matched equities	0.000	2.229	1.000	0.000	-0.000	-0.000	-0.000	0.000
Underlying Markowitz	0.000	2.700	1.000	0.000	-0.000	-0.000	-0.000	0.000

Table 7: Headline factor regressions. The table asks whether option returns are explained by equity drift or volatility state controls before any alpha language is considered; the full eight-underlying regression appears in Appendix A.

The regression is an audit, not a causal proof. If the option book is only a levered large-cap equity book, SPY and single-name betas should explain it. If it is mainly a volatility trade, VX, VIX, and VVIX controls should reveal that. A high point estimate without credible inference, low exposure, and post-cost robustness is not promoted to an option-alpha claim.

Strength	Claim	Boundary
Strong	Option cashflow accounting, Greek-induced covariance, premium-weighted Markowitz formulation, and validation discipline are coherent.	Mathematical construction and reproducibility checks.
Moderate	The point-in-time research implementation produces useful gross and post-cost diagnostics in this sample.	Monthly OPRA Databento-derived panels, exact-settlement rows, and research cost assumptions.
Weak/conditional	Convexity, VIX exposure, and short-premium sleeves help only in selected states after premium, skew, hedge cost, settlement, and risk-budget checks.	State- and settlement-dependent; exact VRO/SQO rows present.
No claim	Live alpha, production tradability, executable capacity, broker-realistic fills, or live margin parity.	No broker-routed orders, live fills, live margin preview, or broker reconciliation.

Table 8: Claim-strength summary. The table distinguishes durable framework claims from sample-dependent diagnostics and from live-trading claims that are not made.

Claim	Type	Evidence	Status
Option covariance equals $B\Omega B^T + \Sigma_\epsilon$	Theorem	Displayed proof and standard covariance algebra	Proved
Unconstrained option tangency is $\Sigma_O^{-1}\mu_O$	Theorem	Displayed Lagrange multiplier proof	Proved
Greek Markowitz Sharpe exceeds random feasible p95	Generated empirical	Performance table and random feasible Sharpe artifact	Not supported after exact-expiry PTT audit
Result is not only long-call equity drift	Generated diagnostic	exposure, regression, regimes, equity benchmarks, leave-one-out	Not supported as an alpha-independence claim
Option premium cost is included in P&L and Sharpe	Accounting identity	NAV weights, mark-return construction, exposure premium columns	Implemented by construction
Empirical returns use point-in-time option choices	Generated diagnostic	timing diagnostics, trading-data audit, and holding-return detail artifacts	Checked against train/test dates
Stock splits do not create option windfalls	Generated diagnostic	raw-close split factors and exact-expiry payoff construction	Checked for detected split/unit jumps
Greek risk model fully explains option P&L	Rejected overclaim	approximation diagnostics and P&L attribution artifacts	Not claimed
Post-cost research returns include implementation frictions	Generated empirical	Generated mid-, half-spread, full-spread, fee, capacity, required-capital, and assignment-risk ledgers	Implemented as conservative research simulation
Pre-production results are broker-executed live evidence	Rejected overclaim	No live fills, order routing, broker margin preview, or broker reconciliation	Not claimed
VIX option results are headline-grade listed-settlement evidence	Conditional empirical	All VIX headline rows use exact VRO/SQO source flags	Supported
Strategy is production tradable after costs	Production claim	No live fills, live broker margin preview, order routing, or broker reconciliation	Not claimed
VIX options are modeled as volatility derivatives, not equity underlyings	Modeling convention	VIX option panel uses VX-forward Black-76 Greeks and VIX/VVIX state variables	Implemented
VIX option expiry P&L is exact listed settlement P&L	Generated empirical	All VIX expiry rows use exact Cboe VRO/SQO settlement	Supported

Table 9: Claim-audit summary. The table separates mathematical claims, empirical diagnostics, settlement-gated VIX claims, post-cost research claims, and production claims that are not claimed.

The claim audit is part of the result. The article can claim a mathematically explicit option-only efficient frontier and a point-in-time empirical implementation. It can report gross, exact-settlement VIX, and pre-production post-cost research diagnostics when the audit layer supports those outputs. It cannot claim production tradability, live fills, live broker margin parity, or executable capacity because those controls are not broker-run in this article.

7. Conclusions

The durable contribution is the option cashflow and risk-accounting framework. A listed option enters the allocation problem only after it has been represented as a premium-funded cashflow, marked under a settlement convention, mapped into delta, gamma, vega, theta, VIX-forward exposure, and residual risk, and forecast under the physical measure. Once those objects exist, the Markowitz algebra is familiar: expected option cashflows and a Greek-induced covariance matrix replace ordinary asset means and covariances.

The validation layer supports a coherent point-in-time research implementation, not production tradability. Gross Sharpe, VIX hedge behavior, and option convexity are not enough. They must survive premium accounting, exact settlement controls, post-cost research account-

ing, beta and Greek attribution, inference, leave-one-out tests, and no-lookahead checks. VIX options are useful in the framework because they are volatility instruments rather than equity substitutes, but their expiry P&L becomes headline-grade only when exact VRO/SOQ settlement supports the rows being reported.

The trader's use case is a disciplined screening rule rather than an automatic trading signal. A sleeve enters the book only when forecasted option premia survive premium cost, spread, settlement, and Greek-budget checks. Convexity is worth paying for only when state variables and hedge diagnostics justify the premium. Short option exposure is rejected or capped when stress loss, assignment risk, liquidity, or margin screens dominate expected carry. VIX options are treated as volatility instruments. When the validation layer cannot separate option premia from equity drift, volatility carry, or optimistic execution, the portfolio should sit in cash collateral, trade smaller, or use simpler diversifiers.

This paper does not claim live-trading performance. The empirical sample is limited to the available OPRA/Databento-derived equity-option panels, VIX option chains, monthly snapshots, and exact VRO/SOQ coverage. Options execution, rates, margin, borrow, assignment, and capacity are modeled through conservative research assumptions rather than broker-level order and fill data. Production deployment would require richer quote histories, executable bid/ask depth, broker margin previews, assignment and exercise modeling, live routing, and reconciliation.

The conclusion is therefore deliberately restrained. Build the option cashflow. Charge the premium and the implementation frictions. Estimate conditional premia before optimizing. Constrain Greeks, beta, VIX exposure, and stress loss. Use regressions, attribution, regimes, leave-one-out tests, and inference to ask whether the result is more than equity drift or volatility carry. Reject any claim that depends on free options, future information, unpriced settlement, optimistic costs, or hidden short-option risk.

Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. Raw licensed OPRA/Databento data are not redistributed. Users with the necessary data rights and API keys can rebuild the local feature panels with the documented data-pull commands and environment template. Exact VRO/SOQ files, when provided locally, are versioned by source hash before VIX option rows become headline-grade.

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A. Appendix

Reproducibility

Run from the project root after configuring local data paths and credentials where needed:

```
make option-paper
make paper
make verify
make test
```

A.1. Additional realized-path and optimizer results

This supplement begins with realized-path and optimizer diagnostics. The exhibits use the same held-out panel as the body tables and compare allocation behavior rather than re-estimating a separate experiment.

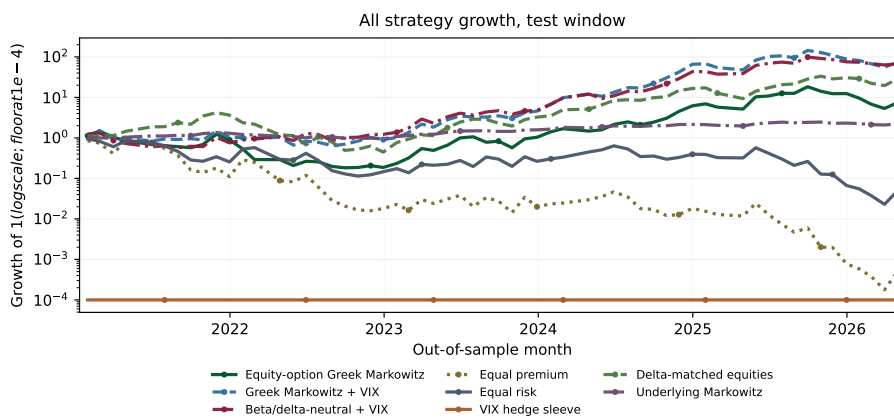


Figure 3: All-strategy out-of-sample growth diagnostic. The main text shows the central strategies; this appendix figure keeps every generated strategy visible for auditability.

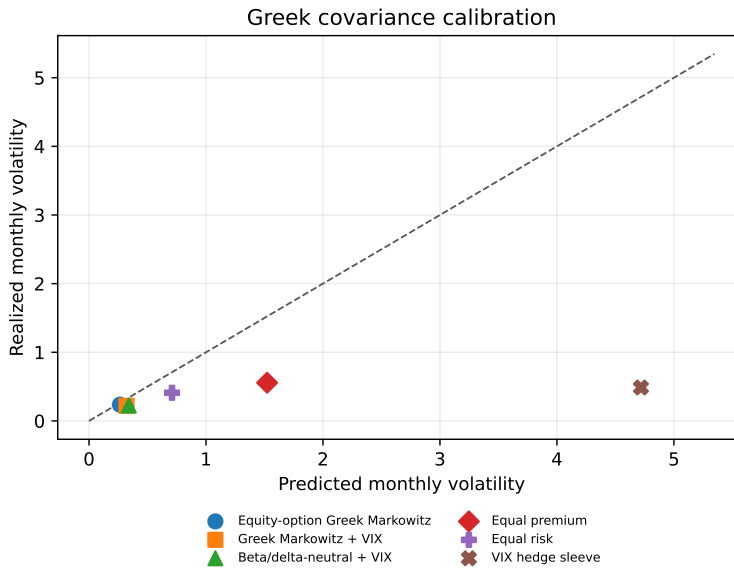


Figure 4: Predicted versus realized monthly volatility by option strategy.

Strategy	Book	Gross NAV	Net NAV	Long premium paid	Short premium sold	Net option premium	Short gross
Equity-option Greek Markowitz	Options	1.000	0.500	0.750	0.250	0.500	0.250
Greek Markowitz + VIX	Options	1.000	0.500	0.750	0.250	0.500	0.250
Beta/delta-neutral + VIX	Options	1.000	0.500	0.750	0.250	0.500	0.250
Equal premium	Options	1.000	1.000	1.000	0.000	1.000	0.000
Equal risk	Options	1.000	1.000	1.000	0.000	1.000	0.000
VIX hedge sleeve	Options	1.000	1.000	1.000	0.000	1.000	0.000
Delta-matched equities	Equities	3.307	3.307				0.000
Underlying Markowitz	Equities	1.000	0.539				0.231

Table 10: Premium and sign exposures. The table records how much option premium is paid, sold, and netted before the strategy return is realized.

Strategy	Book	Net delta	Gross delta	Net gamma	Net vega	Equity vega	VIX vega	Beta SPY proxy	Worst stress return	VIX option gross	Call gross	Put gross
Equity-option Greek Markowitz	Options	2.336	4.055	5.376	0.004	0.004	0.000	3.000	-0.207	0.000	0.682	0.318
Greek Markowitz + VIX	Options	3.307	3.307	5.029	-0.108	0.005	-0.113	2.342	-0.350	0.200	0.633	0.367
Beta/delta-neutral + VIX	Options	-0.086	3.893	2.186	-0.099	0.005	-0.104	-0.145	-0.010	0.200	0.588	0.412
Equal premium	Options	1.208	2.202	359.583	0.211	0.041	0.170	0.897	0.258	0.093	0.537	0.463
Equal risk	Options	0.650	1.988	175.662	0.116	0.027	0.089	0.689	0.126	0.065	0.561	0.439
VIX hedge sleeve	Options	0.000	0.000	18.435	1.572	0.000	1.572	-23.137	-0.406	1.000	1.000	0.000
Delta-matched equities	Equities	3.307	3.307	0.000	0.000	0.000	0.000			0.000		
Underlying Markowitz	Equities	0.539	1.000	0.000	0.000	0.000	0.000			0.000		

Table 11: Greek, beta-proxy, VIX-vega, stress, and option-right exposures.

A.2. Option, VIX, cost, and settlement diagnostics

The option results are reported only after timing, settlement, and cost conventions are visible. This is where the paper records the distinction between gross theory returns, post-cost research returns, and production claims that are not made.

Diagnostic	Value
Return construction	Prior-date option selection, split-adjusted listed-expiry payoff
Minimum option mark filter	0.25
Max train decision date	2020-11-30
Max train realization date	2020-12-31
First test decision date	2020-12-31
First test realization date	2021-01-29
Detected split/unit adjustments	7
Max single option return after repair	48.700
Min single option return after repair	-1.000
Expiry spot source	Raw daily close on listed expiry, or prior trading day if missing
VIX option return construction	Prior-date VIX option selection, VX-forward Greeks, expiry settlement proxy flagged
VIX settlement source	vro_soq_exact:536
VIX first test decision date	2020-12-31

Table 12: Timing and return-construction diagnostics.

Check	Value	Pass
OPRA-derived rows after liquidity filters	160,315	yes
Representative option choices	5,825	yes
Nonmissing option return cells	5,777	yes
Max OPRA spot vs raw close mismatch	2.129e-16	yes
All option choices made before payoff date	True	yes
All payoff dates no later than next rebalance	True	yes
Exact listed-expiry close share	1.000	yes
Max expiry-to-payoff-date lag in days	0	yes
Holding period days min/median/max	15/18.0/21	yes
Rows requiring split conversion during holding	13	yes
Detected raw-close split/unit events	7	yes
Minimum single-option return	-1.000	yes
Maximum single-option return	48.700	yes
Max training decision before first test decision	2020-11-30 before 2020-12-31	yes
Raw VIX option rows after contract parsing	103,650	yes
Representative VIX option choices	630	yes
VIX option holding ledger rows	536	yes
Duplicate VIX date-symbol rows after dedupe	0	yes
Black-76 VX-forward Greek coverage	1.000	yes
VIX settlement source	vro_soq_exact:536	yes
All VIX decisions before payoff dates	True	yes
Minimum VIX long-option return	-1.000	yes

Table 13: Trading-data audit for the point-in-time holding model. Rows include VIX option settlement-source flags; proxy rows, if present, remain diagnostic.

Year	Exact scalar parsed	SOQ component status	RequiredExpiries	FirstExpiration	LastExpiration
2015	False	downloaded	3	2015-12-09	2015-12-30
2015	True	downloaded	10	2015-02-18	2015-11-18
2016	False	unavailable	11	2016-05-04	2016-10-26
2016	True	downloaded	3	2016-01-20	2016-03-16
2016	True	unavailable	4	2016-04-20	2016-12-21
2017	False	unavailable	20	2017-02-15	2017-12-27
2017	True	unavailable	3	2017-01-18	2017-06-21
2018	False	unavailable	6	2018-01-03	2018-10-31
2018	True	unavailable	8	2018-02-14	2018-12-19
2019	True	downloaded	3	2019-10-16	2019-12-18
2019	True	unavailable	8	2019-01-16	2019-09-18
2020	True	downloaded	12	2020-01-22	2020-12-16
2021	True	downloaded	11	2021-01-20	2021-12-22
2022	True	downloaded	10	2022-01-19	2022-12-21
2023	True	downloaded	12	2023-01-18	2023-12-20
2024	True	downloaded	11	2024-01-17	2024-12-18
2025	True	downloaded	11	2025-01-22	2025-12-17
2026	True	downloaded	4	2026-01-21	2026-04-15

Table 14: Public Cboe VRO/SOQ download audit for required VIX expiries. Rows with Exact scalar parsed = False are not used in headline VIX option returns.

Strategy	Mean monthly cost	Ann. cost drag	Spread share	Fee share	Mean margin/NAV	Mean stress margin/NAV	Mean assignment notional/NAV	Capacity penalized share
Beta/delta-neutral + VIX	0.422	5.068	0.099	0.012	4.029	0.235	0.000	0.273
Equal premium	0.259	3.109	0.115	0.025	0.953	0.238	0.000	0.381
Equal risk	0.115	1.381	0.213	0.035	0.920	0.230	0.000	0.347
Equity-option Greek Markowitz	0.733	8.794	0.021	0.003	4.610	0.224	0.039	0.335
Greek Markowitz + VIX	0.610	7.325	0.069	0.008	3.526	0.234	0.000	0.290
VIX hedge sleeve	0.430	5.165	0.347	0.046	0.978	0.244	0.000	0.847

Table 15: Research cost and capital diagnostics. Costs include transaction costs, spread crossing, explicit fees, slippage, borrow, capacity penalties, simulated margin funding, and assignment-risk penalties where flagged. These are not live broker fills.

Strategy	Scenario	Avg contracts traded	Avg quoted spread	Avg monthly cost	Max capacity used	Rejected trades	Mean margin/NAV	Mean stress capital/NAV
Beta/delta-neutral + VIX	full_spread	56.74916809036922	0.03509309915904966	0.0319804537714552	0.9997049345720702	867	1.6454245692301974	0.11046887552661715
Beta/delta-neutral + VIX	half_spread	56.74916809036922	0.03509309915904966	0.01874611132937927	0.9997049345720702	867	1.6454245692301974	0.11046887552661715
Beta/delta-neutral + VIX	mid	56.74916809036922	0.03509309915904966	0.005511768887343023	0.9997049345720702	867	1.6454245692301974	0.11046887552661715
Equal premium	full_spread	65.36655726350517	0.03450510899698828	0.027393513458707507	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal premium	half_spread	65.36655726350517	0.03450510899698828	0.017223026084595215	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal premium	mid	65.36655726350517	0.03450510899698828	0.007052538710482932	0.9993803841618195	1330	0.5895061728395058	0.14737654320987645
Equal risk	full_spread	43.65715485388917	0.03490292057829765	0.0212544284856588	0.9990134907215874	1222	0.5170241047560391	0.12925602618900978
Equal risk	half_spread	43.65715485388917	0.03490292057829765	0.013262382446192645	0.9990134907215874	1222	0.5170241047560391	0.12925602618900978
Equal risk	mid	43.65715485388917	0.03490292057829765	0.005271336643819411	0.9990134907215874	1222	0.5170241047560391	0.12925602618900978
Equity-option Greek Markowitz	full_spread	18.797109490588205	0.017435798276330978	0.006679109598942776	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Equity-option Greek Markowitz	half_spread	18.797109490588205	0.017435798276330978	0.0044518190699674685	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Equity-option Greek Markowitz	mid	18.797109490588205	0.017435798276330978	0.002224528540992161	0.9901002345178692	957	1.3950575743846914	0.07960287865721945
Greek Markowitz + VIX	full_spread	49.294675424203284	0.035957144195152634	0.026579996362664688	0.9988451122012506	898	1.5123535571133067	0.10149951553572087
Greek Markowitz + VIX	half_spread	49.294675424203284	0.035957144195152634	0.01563456341722021	0.9988451122012506	898	1.5123535571133067	0.10149951553572087
Greek Markowitz + VIX	mid	49.294675424203284	0.035957144195152634	0.0046891304717757345	0.9988451122012506	898	1.5123535571133067	0.10149951553572087
VIX hedge sleeve	full_spread	2410.1807707861826	0.15	0.07580327078625228	0.988507805945911	153	0.14999999999999997	0.03749999999999999
VIX hedge sleeve	half_spread	2410.1807707861826	0.15	0.042053270786252275	0.988507805945911	153	0.14999999999999997	0.03749999999999999
VIX hedge sleeve	mid	2410.1807707861826	0.15	0.008303270786252284	0.988507805945911	153	0.14999999999999997	0.03749999999999999

Table 16: Executable-cost, capacity, and market-impact diagnostics by scenario. Rows are generated from the cost-scenario ledger and record no-fill/capacity failures rather than assuming midpoint execution.

Strategy	Downside ann. dev	Max drawdown	Worst month	SR 90% CI lo	SR 90% CI hi	Pred./realized vol	Gross NAV	Net NAV
Equity-option Greek Markowitz	0.505	-0.876	-0.455	0.842	0.083	1.089	1.000	0.500
Greek Markowitz + VIX	0.368	-0.613	-0.371	1.374	0.602	1.385	1.000	0.500
Beta/delta-neutral + VIX	0.344	-0.570	-0.354	1.414	0.676	1.478	1.000	0.500
Equal premium	1.004	-1.000	-0.662	-0.030	-0.635	2.782	1.000	1.000
Equal risk	0.713	-0.976	-0.516	0.082	-0.558	1.814	1.000	1.000
VIX hedge sleeve	3.006	-1.000	-1.000	-6.429	-9.004	11.150	1.000	1.000
Delta-matched equities	0.473	-0.894	-0.454	1.176	0.372	3.307	3.307	3.307
Underlying Markowitz	0.096	-0.303	-0.073	0.909	0.103	1.000	1.000	0.539

Table 17: Gross performance diagnostics and NAV accounting.

Strategy	Downside ann. dev	Max drawdown	Worst month	SR 90% CI lo	SR 90% CI hi	Pred./realized vol	Gross NAV	Net NAV
Equity-option Greek Markowitz	5.864	-1.243	-10.800	-2.209	-1.164		1.000	0.500
Greek Markowitz + VIX	5.074	-1.183	-9.226	-1.871	-0.994		1.000	0.500
Beta/delta-neutral + VIX	2.997	-1.000	-5.540	-1.961	-1.094		1.000	0.500
Equal premium	2.107	-1.037	-2.431	-1.975	-0.816		1.000	1.000
Equal risk	1.065	-1.000	-0.974	-1.576	-0.486		1.000	1.000
VIX hedge sleeve	5.876	-1.159	-9.482	-7.714	-2.676		1.000	1.000
Delta-matched equities	0.473	-0.894	-0.454	0.372	2.182			
Underlying Markowitz	0.096	-0.303	-0.073	0.103	1.812			

Table 18: Post-cost performance diagnostics and NAV accounting.

A.3. Validation, inference, and robustness additional results

The validation appendix keeps uncertain evidence in view. The bootstrap intervals and leave-one-underlying-out tests are uncertainty screens, not proof that a premium is stable.

Liquidity tier	Strategy	Ann. return	Ann. vol	Sharpe	Calmar	Omega
all_eligible	Greek Markowitz	1.2463192285080007	0.9068172193093107	1.3743885779509968	2.033918612174948	2.7843631987725814
all_eligible	Equal premium	-0.05552159864526898	1.878473564887012	-0.02955676336526383	-0.05553271091792252	0.9771765473237329
all_eligible	Equal risk	0.10434553413679803	1.2751366252116842	0.0818308658646486	0.10693488072634062	1.0641387065829713
all_eligible	Delta neutral	1.1720618697652665	0.8291774074138495	1.4135236431740843	2.057353788525215	2.947600334500001
all_eligible	VIX hedge sleeve	-9.28491238522107	1.444181867492721	-6.429184989935395	nan	0.04042569095025844
tight_spread_quartile	Greek Markowitz	0.7489053854294916	0.9298739055330015	0.8053838063132019	1.0252766141786784	1.9114377746808244
tight_spread_quartile	Equal premium	1.9315962727008187	2.264282965233791	0.8530719447873328	1.9570493784559104	1.9570538593238194
tight_spread_quartile	Equal risk	1.1814619447081522	1.6060721114200187	0.7356219788061417	1.2213787266592875	1.731852284214156
tight_spread_quartile	Delta neutral	0.6871581895298814	0.8626278440164994	0.7965870732045813	0.9516452083830749	1.8586123893126145
top_volume_quartile	Greek Markowitz	2.0376224599150063	1.577048405020137	1.2920481409630469	2.7131456082828156	2.672187767038167
top_volume_quartile	Equal premium	-0.9862571402804206	2.1637646209976955	-0.45580611250851527	-0.9862571478228792	0.710630732496604
top_volume_quartile	Equal risk	-0.9851204806746627	1.6783050234078585	-0.5869734445972999	-0.9851214694043257	0.6612429674757221
top_volume_quartile	Delta neutral	1.8907359027828463	1.4349238472989643	1.3176559198676352	2.475447246701459	2.6359199548389416
top_volume_quartile	VIX hedge sleeve	-9.28491238522107	1.444181867492721	-6.429184989935395	nan	0.04042569095025844

Table 19: Liquidity-tier performance. The tiers test whether the option allocation survives when the universe is restricted to high-volume, tight-spread, high-open-interest, or jointly liquid contracts.

Strategy	Equity delta	Equity gamma	Equity vega	VIX-forward delta	VIX-forward gamma	VIX-option vega	Theta/carry	VX roll	Skew/tail	Residual	Realized ann. mean	Residual vol share
Equity-option Greek Markowitz	0.521	0.898	0.002	0.000	0.000	0.000	-0.003	0.000	0.000	-0.524	0.894	0.570
Greek Markowitz + VIX	0.489	1.475	0.001	0.767	-0.867	-0.227	1.057	0.000	0.000	-1.447	1.246	0.950
Beta/delta-neutral + VIX	0.168	1.486	0.001	0.554	-0.745	-0.190	1.034	0.000	0.000	-1.135	1.172	1.069
Equal premium	0.062	7.928	0.006	-0.016	0.358	0.045	-1.016	0.000	0.000	-7.423	-0.056	0.354
Equal risk	0.027	5.625	0.004	-0.096	0.270	0.049	-0.592	0.000	0.000	-5.184	0.104	0.514
VIX hedge sleeve	0.000	0.000	0.000	2.083	2.615	0.135	-12.254	0.000	0.000	-1.865	-9.285	4.858

Table 20: Annualized mean P&L attribution. Equity-option Greeks and VIX-forward Greeks are separated; components reconcile to realized mean after the residual term.

Strategy	Regime	Months	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio
Equity-option Greek Markowitz	Down	19	-0.683	0.847	-0.806	-1.975	0.104	-1.017	-0.811	0.572	-0.225
Equity-option Greek Markowitz	Flat	21	0.518	0.928	0.558	-0.761	1.556	1.137	0.823	1.525	-0.335
Equity-option Greek Markowitz	Up	20	2.787	1.175	2.373	0.685	4.553	8.069	6.615	6.928	1.808
Greek Markowitz + VIX	Down	19	0.806	0.746	1.081	0.153	1.938	2.583	1.918	2.235	1.764
Greek Markowitz + VIX	Flat	21	0.562	0.917	0.612	-0.991	1.786	1.343	0.965	1.618	0.073
Greek Markowitz + VIX	Up	20	2.384	0.985	2.421	0.831	4.232	6.598	7.182	5.575	1.737
Beta/delta-neutral + VIX	Down	19	1.487	0.721	2.061	1.366	3.248	4.939	5.123	4.422	2.700
Beta/delta-neutral + VIX	Flat	21	0.649	0.815	0.796	-0.666	1.860	2.154	1.757	1.990	0.151
Beta/delta-neutral + VIX	Up	20	1.423	0.951	1.496	-0.182	3.218	3.419	3.357	3.019	0.777
Equal premium	Down	19	-0.515	1.890	-0.273	-1.562	0.450	-0.473	-0.535	0.812	-0.013
Equal premium	Flat	21	-2.368	0.879	-2.694	-5.258	-1.256	-2.270	-2.377	0.155	-3.484
Equal premium	Up	20	2.809	2.352	1.195	0.314	2.010	3.236	3.712	2.597	0.917
Equal risk	Down	19	0.101	1.436	0.070	-0.954	0.868	0.128	0.130	1.054	0.411
Equal risk	Flat	21	-1.654	0.687	-2.409	-4.471	-0.990	-2.113	-1.703	0.155	-3.447
Equal risk	Up	20	1.954	1.419	1.377	0.302	2.376	3.665	3.495	2.843	0.915
VIX hedge sleeve	Down	19	-7.743	1.368	-5.662	-9.800	-4.100	-2.982		0.021	-5.203
VIX hedge sleeve	Flat	21	-8.367	1.909	-4.383	-6.184	-3.226	-2.829		0.104	-4.129
VIX hedge sleeve	Up	20	-11.714	0.276	-42.492	-107.213	-26.949	-3.453		0.000	-40.570
Delta-matched equities	Down	19	-1.767	0.675	-2.619	-4.396	-1.375	-2.204	-1.817	0.159	-2.086
Delta-matched equities	Flat	21	1.272	0.707	1.800	0.747	2.788	5.921	3.839	5.091	1.172
Delta-matched equities	Up	20	3.630	0.698	5.203	3.511	8.613	34.716	26.891	45.819	4.428
Underlying Markowitz	Down	19	-0.319	0.120	-2.664	-4.820	-1.265	-2.262	-0.765	0.148	1.948
Underlying Markowitz	Flat	21	0.172	0.185	0.933	-0.496	2.168	2.087	1.120	2.038	-0.742
Underlying Markowitz	Up	20	0.658	0.172	3.829	2.463	6.121	15.942	17.455	15.149	-0.128

Table 21: Performance by SPY-return tercile. Down, flat, and up regimes are descriptive cuts of the held-out sample, not fitted optimizer inputs.

Ablation	Ann. return	Ann. vol	Sharpe	Calmar	Omega	Active assets
carry_only	6.70147099899699	1.9164807271628144	3.496758878925928	2.8096258939800632e-05	8.037348345976534	54
variance_risk_premium_only	9.63139175996257	0.6700827342459895	14.373436693306674	nan	nan	54
skew_tail_only	4.489045206711676	3.2048741838250314	1.4006931159319276	0.003079353839745213	2.691677332065947	18
vix_regime_only	4.590815059335688	2.544245648201727	1.8043914362516356	0.004298213073203248	3.3862924961758027	54
relative_value_only	2.5789118807598546	1.4076551542781681	1.8320622582327668	7.262838528939612	6.69566248471262	54
full_conditional_model	4.113956564258069	2.9681846972242805	1.3860177124776856	2.1877214234861713	3.1853840590190665	54

Table 22: Forecast ablation performance. Each row isolates one economically identified component of the conditional option-premium forecast.

Strategy	Metric	Estimate	CI lo	CI hi	N	Method	Block size	Seed	Return basis
Equity-option Greek Markowitz	Sharpe	0.842	0.083	1.679	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Sortino	1.770	0.141	4.581	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Calmar	1.020	0.083	2.828	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Omega	1.888	1.064	3.546	60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Information Ratio	0.693			60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Sharpe	1.374	0.602	2.222	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Sortino	3.385	1.188	7.275	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Calmar	2.034	0.700	4.901	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Omega	2.784	1.562	5.251	60	circular block bootstrap	8	20260625	Gross before costs
Greek Markowitz + VIX	Information Ratio	1.192			60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Sharpe	1.414	0.676	2.220	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Sortino	3.410	1.335	6.541	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Calmar	2.057	0.701	5.611	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Omega	2.948	1.693	5.333	60	circular block bootstrap	8	20260625	Gross before costs
Beta/delta-neutral + VIX	Information Ratio	1.219			60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Sharpe	-0.030	-0.635	0.437	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Sortino	-0.055	-0.946	0.920	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Calmar	-0.056	-1.051	0.849	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Omega	0.977	0.624	1.416	60	circular block bootstrap	8	20260625	Gross before costs
Equal premium	Information Ratio	-0.260			60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Sharpe	0.082	-0.558	0.627	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Sortino	0.146	-0.821	1.305	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Calmar	0.107	-0.667	0.954	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Omega	1.064	0.662	1.621	60	circular block bootstrap	8	20260625	Gross before costs
Equal risk	Information Ratio	-0.151			60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Sharpe	-6.429	-9.004	-4.921	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Sortino	-3.089	-3.253	-2.904	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Calmar		-10.058	-8.262	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Omega	0.040	0.016	0.071	60	circular block bootstrap	8	20260625	Gross before costs
VIX hedge sleeve	Information Ratio	-6.344			60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Sharpe	1.176	0.372	2.182	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Sortino	2.317	0.615	5.886	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Calmar	1.225	0.388	3.716	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Omega	2.367	1.301	5.182	60	circular block bootstrap	8	20260625	Gross before costs
Delta-matched equities	Information Ratio	0.995			60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Sharpe	0.909	0.103	1.812	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Sortino	1.856	0.172	4.958	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Calmar	0.590	0.045	2.544	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Omega	1.929	1.074	3.877	60	circular block bootstrap	8	20260625	Gross before costs
Underlying Markowitz	Information Ratio	0.166			60	circular block bootstrap	8	20260625	Gross before costs
Equity-option Greek Markowitz	Sharpe	-1.443	-2.209	-1.164	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Sortino	-1.347	-1.895	-1.123	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Calmar	-6.357	-11.009	-1.856	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Omega	0.098	0.045	0.218	60	circular block bootstrap	8	20260625	Post-cost research
Equity-option Greek Markowitz	Information Ratio	-1.492			60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Sharpe	-1.259	-1.871	-0.994	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Sortino	-1.198	-1.683	-0.971	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Calmar	-5.136	-8.117	-1.087	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Omega	0.116	0.054	0.269	60	circular block bootstrap	8	20260625	Post-cost research
Greek Markowitz + VIX	Information Ratio	-1.303			60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Sharpe	-1.371	-1.961	-1.094	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Sortino	-1.300	-1.790	-1.073	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Calmar	-3.895	-6.033	-1.289	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Omega	0.151	0.075	0.321	60	circular block bootstrap	8	20260625	Post-cost research
Beta/delta-neutral + VIX	Information Ratio	-1.418			60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Sharpe	-1.331	-1.975	-0.816	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Sortino	-1.502	-1.923	-1.066	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Calmar	-3.051	-4.189	-1.543	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Omega	0.359	0.206	0.547	60	circular block bootstrap	8	20260625	Post-cost research
Equal premium	Information Ratio	-1.619			60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Sharpe	-0.934	-1.576	-0.486	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Sortino	-1.199	-1.714	-0.697	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Calmar	-1.276	-1.962	-0.685	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Omega	0.505	0.326	0.702	60	circular block bootstrap	8	20260625	Post-cost research
Equal risk	Information Ratio	-1.225			60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Sharpe	-3.413	-7.714	-2.676	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Sortino	-2.445	-3.171	-2.131	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Calmar	-12.391	-15.461	-7.145	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Omega	0.016	0.004	0.032	60	circular block bootstrap	8	20260625	Post-cost research
VIX hedge sleeve	Information Ratio	-3.402			60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Sharpe	1.176	0.372	2.182	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Sortino	2.317	0.615	5.886	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Calmar	1.225	0.388	3.716	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Omega	2.367	1.301	5.182	60	circular block bootstrap	8	20260625	Post-cost research
Delta-matched equities	Information Ratio	0.995			60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Sharpe	0.909	0.103	1.812	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Sortino	1.856	0.172	4.958	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Calmar	0.590	0.045	2.544	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Omega	1.929	1.074	3.877	60	circular block bootstrap	8	20260625	Post-cost research
Underlying Markowitz	Information Ratio	0.166			60	circular block bootstrap	8	20260625	Post-cost research

Table 23: Full bootstrap inference ledger for gross and post-cost strategy metrics.

Variant	Sharpe	Probabilistic Sharpe	Deflated Sharpe	Bootstrap max Sharpe p95	N	Block size	Seed
Equity-option Greek Markowitz: gross	0.8421194565895301	0.9999999999997874	0.0	2.395856192979967	60	8	20260625
Greek Markowitz + VIX: gross	1.3743885779509968	1.0	1.171660871221686e-190	2.395856192979967	60	8	20260625
Beta/delta-neutral + VIX: gross	1.4135236431740843	1.0	5.5953905332384846e-164	2.395856192979967	60	8	20260625
Equal premium: gross	-0.02955676336526383	0.4122235847317047	0.0	2.395856192979967	60	8	20260625
Equal risk: gross	0.0818308658646486	0.7451516479581755	0.0	2.395856192979967	60	8	20260625
VIX hedge sleeve: gross	-6.429184989935395	1.1855878159463198e-08	9.87288524527083e-25	2.395856192979967	60	8	20260625
Delta-matched equities: gross	1.1763702582317712	0.9999999999999847	7.411543205687691e-158	2.395856192979967	60	8	20260625
Underlying Markowitz: gross	0.9088833301008382	0.9999999999999218	3.601643297065905e-281	2.395856192979967	60	8	20260625
Equity-option Greek Markowitz: mid	0.8170505717670248	0.9999999999992117	0.0	2.395856192979967	60	8	20260625
Greek Markowitz + VIX: mid	1.3138775407676593	1.0	2.0832065107842354e-203	2.395856192979967	60	8	20260625
Beta/delta-neutral + VIX: mid	1.3352567190175022	1.0	8.293236346779767e-180	2.395856192979967	60	8	20260625
Equal premium: mid	-0.07459766125108384	0.29440021003850236	0.0	2.395856192979967	60	8	20260625
Equal risk: mid	0.032216351568761376	0.5995571041200423	0.0	2.395856192979967	60	8	20260625
VIX hedge sleeve: mid	-6.4606628766839735	1.0514523448932735e-08	8.416119683516885e-25	2.395856192979967	60	8	20260625
Equity-option Greek Markowitz: half_spread	0.7919362861614629	0.9999999999996282	0.0	2.395856192979967	60	8	20260625
Greek Markowitz + VIX: half_spread	1.170321552235334	1.0	2.6433421736216467e-233	2.395856192979967	60	8	20260625
Beta/delta-neutral + VIX: half_spread	1.143232107598715	1.0	4.726688767518834e-220	2.395856192979967	60	8	20260625
Equal premium: half_spread	-0.13959427042490616	0.16788800945156535	2.979791317633e-310	2.395856192979967	60	8	20260625
Equal risk: half_spread	-0.042972181890083776	0.37381543239837334	0.0	2.395856192979967	60	8	20260625
VIX hedge sleeve: half_spread	-6.540853900526196	7.56882120672634e-09	5.219977061164737e-25	2.395856192979967	60	8	20260625
Equity-option Greek Markowitz: full_spread	0.7668067218645858	0.9999999999988043	0.0	2.395856192979967	60	8	20260625
Greek Markowitz + VIX: full_spread	1.0258219183556512	0.9999999999999999	4.061857790130324e-263	2.395856192979967	60	8	20260625
Beta/delta-neutral + VIX: full_spread	0.9508576292459209	0.999999999999716	2.0390393622011163e-261	2.395856192979967	60	8	20260625
Equal premium: full_spread	-0.20462197552298175	0.08945368147995542	1.711455237880309e-288	2.395856192979967	60	8	20260625
Equal risk: full_spread	-0.11815498556113241	0.19879777872024146	0.0	2.395856192979967	60	8	20260625
VIX hedge sleeve: full_spread	-6.607601788466617	4.860837695911706e-09	2.0508103641611321e-25	2.395856192979967	60	8	20260625

Table 24: Reality-check inference across gross and executable-cost variants.

A.3.1 Tail-path simulation diagnostics

The next tables simulate alternative out-of-sample paths from the realized monthly strategy returns. They are path-risk diagnostics only: simulated returns do not enter contract selection, optimizer weights, expected-return forecasts, or any trading decision. The block bootstrap preserves short-run dependence in the realized option return path. The volatility-clustered simulation fits EGARCH(1,1)-t only when the OOS sample is long enough; otherwise it reports an EWMA residual fallback rather than pretending that the EGARCH parameters are precisely estimated.

Return basis	Strategy	Requested method	Simulation	Breach 10%	Breach 25%	Breach 50%	Breach 75%	Breach 90%
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.983	0.770	0.298
Gross before costs	Equity-option Greek Markowitz	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.987	0.708	0.296
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	0.996	0.795	0.135	0.005
Gross before costs	Greek Markowitz + VIX	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	0.999	0.884	0.379	0.136
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	0.996	0.585	0.063	0.001
Gross before costs	Beta/delta-neutral + VIX	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	0.996	0.671	0.176	0.037
Gross before costs	Delta-matched equities	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.902	0.548	0.175
Gross before costs	Delta-matched equities	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.896	0.358	0.069
Gross before costs	Underlying Markowitz	circular_block_bootstrap	circular_block_bootstrap	0.981	0.432	0.013	0.000	0.000
Gross before costs	Underlying Markowitz	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	0.981	0.272	0.002	0.000	0.000
Gross before costs	Equal premium	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.999	0.995
Gross before costs	Equal premium	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	0.986	0.986	0.986	0.986	0.985
Gross before costs	Equal risk	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.973	0.841
Gross before costs	Equal risk	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	0.993	0.937
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	circular_block_bootstrap	0.437	0.437	0.437	0.437	0.437
Gross before costs	VIX hedge sleeve	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	1.000	1.000
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.986	0.789	0.343
Full-spread post-cost	Equity-option Greek Markowitz	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.993	0.756	0.359
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	0.997	0.913	0.374	0.060
Full-spread post-cost	Greek Markowitz + VIX	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.948	0.503	0.181
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.799	0.251	0.035
Full-spread post-cost	Beta/delta-neutral + VIX	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.850	0.295	0.068
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	0.902	0.548	0.175
Full-spread post-cost	Delta-matched equities	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	0.896	0.358	0.069
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	circular_block_bootstrap	0.981	0.432	0.013	0.000	0.000
Full-spread post-cost	Underlying Markowitz	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	0.981	0.272	0.002	0.000	0.000
Full-spread post-cost	Equal premium	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	1.000	0.999
Full-spread post-cost	Equal premium	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	0.986	0.986	0.986	0.986	0.986
Full-spread post-cost	Equal risk	circular_block_bootstrap	circular_block_bootstrap	1.000	1.000	1.000	0.989	0.938
Full-spread post-cost	Equal risk	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	1.000	0.984
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	circular_block_bootstrap	0.471	0.471	0.471	0.471	0.471
Full-spread post-cost	VIX hedge sleeve	egarch_or_ewma	ewma_residual_fallback_insufficient_egarch_obs	1.000	1.000	1.000	1.000	1.000

Table 25: Drawdown-breach probabilities from block-bootstrap and volatility-clustered path simulations. The thresholds are intentionally wide because option-only post-cost paths can experience large drawdowns.

Return basis	Strategy	Simulation	N paths	Ann. return p50	Sortino p50	Max 100 p50	Max 100 p90	Terminal wealth p50
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	1000	-0.212	1.626	-0.976	-0.880	0.128
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	1000	-0.176	1.861	-0.84	-0.819	0.128
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	1000	1.382	1.451	-0.818	-0.802	1.023
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	1000	0.802	1.501	-0.801	-0.792	1.000
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	1.124	1.280	-0.777	-0.747	4.501
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	1.274	1.238	-0.878	-0.771	2.717
Gross before costs	Delta-matched equities	circular_block_bootstrap	1000	1.008	2.410	-0.804	-0.774	0.717
Gross before costs	Delta-matched equities	circular_block_bootstrap	1000	1.281	2.841	-0.819	-0.800	1.109
Gross before costs	Underlying Markowitz	circular_block_bootstrap	1000	0.999	1.491	-0.819	-0.777	0.999
Gross before costs	Underlying Markowitz	circular_block_bootstrap	1000	0.149	1.491	-0.832	-0.819	1.006
Gross before costs	Equal premium	circular_block_bootstrap	1000	0.784	1.004	-1.000	-1.000	0.801
Gross before costs	Equal premium	circular_block_bootstrap	1000	-0.016	-0.012	-1.000	-1.000	-0.000
Gross before costs	Equal risk	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	1000	-0.126	1.151	-0.980	-0.877	0.096
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	1000	0.413	1.626	-0.982	-0.885	0.096
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	1000	0.613	1.426	-0.980	-0.885	0.003
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	1000	0.727	1.171	-0.977	-0.719	0.014
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	0.913	2.263	-0.980	-0.885	0.003
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	1000	0.727	1.171	-0.977	-0.719	0.014
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	1000	0.810	2.702	-0.919	-0.871	0.014
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	1000	0.176	1.810	-0.822	-0.751	0.014
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	1000	1.281	2.410	-0.819	-0.800	1.006
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	1000	0.176	1.810	-0.822	-0.751	0.014
Full-spread post-cost	Equal premium	circular_block_bootstrap	1000	0.149	1.491	-0.832	-0.819	1.006
Full-spread post-cost	Equal premium	circular_block_bootstrap	1000	-0.016	-0.012	-1.000	-1.000	-0.000
Full-spread post-cost	Equal risk	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	1000	-0.008	-0.002	-1.000	-1.000	-0.000

Table 26: Tail-path simulation summary. Median return and Sortino describe the simulated path center; the fifth-percentile maximum drawdown and terminal wealth columns describe the left tail of realized-path resampling.

Return basis	Strategy	Method	Status	Reason	N obs	Source start	Source end	Block length	Path count	Lag 1 autocorr
Gross before costs	Equity-option Greek Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.096
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.096
Gross before costs	Greek Markowitz + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.003
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.003
Gross before costs	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.083
Gross before costs	Delta-matched equities	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.083
Gross before costs	Delta-matched equities	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.014
Gross before costs	Underlying Markowitz	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.014
Gross before costs	Underlying Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.041
Gross before costs	Equal premium	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.041
Gross before costs	Equal premium	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.256
Gross before costs	Equal risk	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.256
Gross before costs	Equal risk	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.188
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.188
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.125
Gross before costs	VIX hedge sleeve	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.125
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.096
Full-spread post-cost	Equity-option Greek Markowitz	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.096
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.008
Full-spread post-cost	Greek Markowitz + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.008
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.096
Full-spread post-cost	Beta/delta-neutral + VIX	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.096
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.014
Full-spread post-cost	Delta-matched equities	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.014
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	0.001
Full-spread post-cost	Underlying Markowitz	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	0.041
Full-spread post-cost	Equal premium	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.256
Full-spread post-cost	Equal premium	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.256
Full-spread post-cost	Equal risk	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.188
Full-spread post-cost	Equal risk	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.188
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	ok	insufficient_egarch_obs	60	2021-01-29	2026-04-30	6.000	1000	-0.107
Full-spread post-cost	VIX hedge sleeve	circular_block_bootstrap	ok	nan	60	2021-01-29	2026-04-30	6.000	1000	-0.107

Table 27: Simulation assumptions and model status. Fallback rows are deliberate diagnostics: when the monthly OOS sample is too short for EGARCH estimation, the appendix uses an EWMA residual bootstrap and records the reason.

Strategy	Regime	Months	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio
Equity-option Greek Markowitz	Low VIX	20	1.585	0.919	1.724	1.033	2.559	4.862	5.911	3.697	1.954
Equity-option Greek Markowitz	Mid VIX	20	1.104	1.216	0.907	-0.388	1.845	2.157	2.351	2.017	0.841
Equity-option Greek Markowitz	High VIX	20	-0.006	1.029	-0.006	-1.030	1.097	-0.009	-0.007	0.996	-0.808
Greek Markowitz + VIX	Low VIX	20	1.386	0.846	1.638	0.265	3.472	4.774	4.419	3.257	2.330
Greek Markowitz + VIX	Mid VIX	20	1.533	1.054	1.455	0.412	2.331	3.812	4.985	3.198	1.408
Greek Markowitz + VIX	High VIX	20	0.821	0.838	0.979	-0.096	2.111	2.047	2.057	2.047	-0.223
Beta/delta-neutral + VIX	Low VIX	20	1.176	0.761	1.546	0.252	3.145	3.811	5.194	2.948	2.396
Beta/delta-neutral + VIX	Mid VIX	20	0.974	0.965	1.009	-0.446	2.152	2.326	2.350	2.250	1.028
Beta/delta-neutral + VIX	High VIX	20	1.366	0.786	1.737	0.804	2.874	4.716	4.063	4.231	0.383
Equal premium	Low VIX	20	0.249	1.592	0.157	-1.511	1.123	0.283	0.286	1.132	0.591
Equal premium	Mid VIX	20	0.944	2.287	0.413	-0.680	1.136	0.923	1.004	1.371	0.358
Equal premium	High VIX	20	-1.360	1.717	-0.792	-3.832	0.263	-1.240	-1.372	0.524	-1.345
Equal risk	Low VIX	20	0.598	1.122	0.533	-0.926	1.641	0.983	0.824	1.488	1.038
Equal risk	Mid VIX	20	0.348	1.456	0.239	-0.779	0.996	0.488	0.436	1.204	0.231
Equal risk	High VIX	20	-0.633	1.264	-0.501	-2.543	0.500	-0.787	-0.679	0.676	-1.227
VIX hedge sleeve	Low VIX	20	-9.290	1.621	-5.730	-20.577	-3.438	-3.036	-9.290	0.063	-5.189
VIX hedge sleeve	Mid VIX	20	-10.031	1.331	-7.534	-41.311	-4.785	-3.203	0.038	-7.879	-7.879
VIX hedge sleeve	High VIX	20	-8.533	1.406	-6.069	-9.076	-4.610	-3.031	0.018	-7.261	-7.261
Delta-matched equities	Low VIX	20	2.273	0.692	3.283	2.654	4.097	19.209	15.977	17.371	2.903
Delta-matched equities	Mid VIX	20	1.918	0.917	2.091	1.374	2.983	6.416	6.081	5.040	1.689
Delta-matched equities	High VIX	20	-0.904	0.917	-0.986	-2.049	-0.174	-1.201	-0.980	0.495	-1.403
Underlying Markowitz	Low VIX	20	0.350	0.174	2.008	0.968	3.220	5.384	4.524	4.537	1.879
Underlying Markowitz	Mid VIX	20	0.386	0.194	1.994	1.369	2.695	7.002	7.409	5.008	0.746
Underlying Markowitz	High VIX	20	-0.201	0.181	-1.108	-2.171	-0.324	-1.403	-0.622	0.473	-1.495

Table 28: Performance by VIX-level tercile.

Strategy	Ann. alpha	Alpha HAC t	Alpha HAC se	HAC lags	R ²	Residual ann. vol	N	Beta SPY	Beta AAPL	Beta AMZN	Beta GOOGL	Beta JPM	Beta META	Beta MSFT	Beta NVDA	Beta TSLA	Beta VX front	Beta dVIX	Beta dVVIX
Equity-option Greek Markowitz	0.405	1.141	0.355	3	0.618	0.634	58	0.005	-1.331	-0.875	0.631	0.590	0.126	2.585	-0.089	1.136	-1.368	0.046	0.003
Greek Markowitz + VIX	0.675	2.060	0.328	3	0.513	0.616	58	1.908	-1.462	-0.747	0.625	-0.127	-0.026	1.877	0.222	0.871	-0.767	0.056	-0.001
Beta delta neutral + VIX	0.845	2.753	0.307	3	0.429	0.601	58	0.715	-1.475	-0.612	0.472	-0.579	-0.170	1.336	0.370	0.815	-0.522	0.038	-0.001
Equal premium	-0.416	-0.566	0.735	3	0.253	1.647	58	2.522	-1.714	-0.069	1.943	-1.879	-0.344	3.001	-0.374	0.404	-0.749	0.032	-0.001
Equal risk	-0.183	-0.334	0.547	3	0.253	1.115	58	2.041	-1.795	-0.113	1.331	-1.484	-0.316	1.968	-0.082	0.321	-0.379	0.020	-0.003
VIX hedge sleeve	-8.807	-14.623	0.602	3	0.252	1.264	58	3.562	-0.597	-0.589	0.670	-3.047	0.423	-0.958	-0.030	-0.131	-1.180	0.093	-0.013
Delta-matched equities	0.000	2.229	0.000	3	1.000	0.000	58	-0.000	0.088	0.539	0.230	0.590	0.011	0.442	0.778	0.628	-0.000	-0.000	0.000
Underlying Markowitz	0.000	2.700	0.000	3	1.000	0.000	58	-0.000	-0.085	0.159	-0.096	-0.050	0.029	0.280	0.191	0.111	-0.000	-0.000	0.000

Table 29: Full factor regression ledger, including SPY, all eight underlying equities, VX front returns, VIX changes, and VVIX changes.

Exclusion	Remaining underlyings	Option assets	Ann. return	Ann. vol	Sharpe	Sharpe CI lo	Sharpe CI hi	Sortino	Calmar	Omega	Info. ratio	Net delta	Beta SPY
All underlyings	8	54	1.246	0.907	1.374	0.601	2.209	3.385	2.034	2.784	1.200	3.307	1.297
No AAPL	7	48	1.253	0.949	1.321	0.571	2.091	3.233	2.016	2.673	1.147	3.216	1.240
No AMZN	7	48	1.286	0.920	1.397	0.578	2.242	3.710	2.097	2.899	1.223	3.344	1.178
No GOOGL	7	49	1.214	0.895	1.357	0.547	2.213	3.325	1.936	2.740	1.180	3.246	1.265
No JPM	7	49	1.202	0.917	1.311	0.650	2.015	3.032	2.540	2.741	1.106	3.186	1.105
No META	7	47	1.250	0.944	1.323	0.566	2.149	3.188	2.011	2.658	1.163	3.381	1.375
No MSFT	7	49	1.410	0.955	1.476	0.750	2.228	3.807	2.386	3.053	1.312	3.406	1.627
No NVDA	7	46	1.085	0.821	1.322	0.547	2.151	3.256	1.893	2.750	1.157	3.571	0.775
No TSLA	7	47	1.198	0.842	1.424	0.671	2.251	3.612	1.988	3.118	1.265	3.330	1.287
No META/NVDA/TSLA	5	32	0.934	0.837	1.115	0.336	1.940	2.279	1.407	2.352	0.968	4.019	1.217

Table 30: Leave-one-underlying-out robustness for the VIX-enabled Greek Markowitz option portfolio.

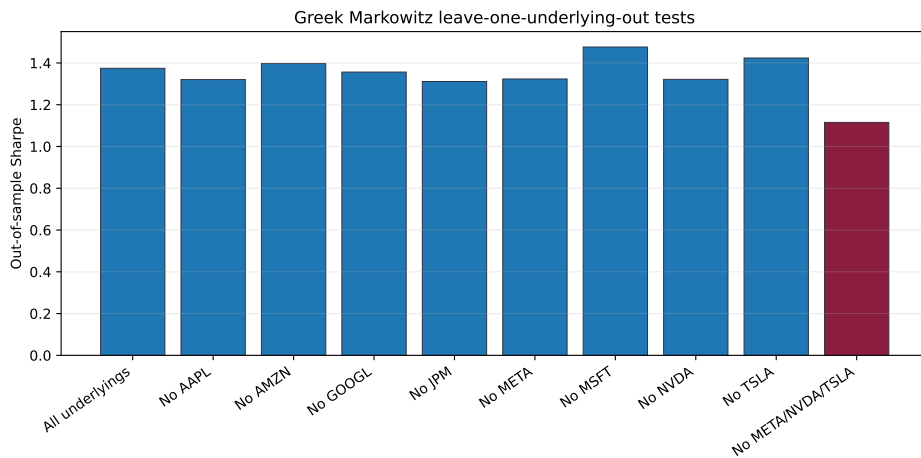


Figure 5: Out-of-sample Sharpe ratios for leave-one-underlying-out variants.

Diagnostic	Value
Rolling 36M OOS months	20.000
Rolling 36M OOS ann. return	1.460
Rolling 36M OOS ann. vol	0.882
Rolling 36M OOS Sharpe	1.656
Rolling 36M OOS Sortino	2.983
Rolling 36M OOS Calmar	2.561
Rolling 36M OOS Omega	3.307

Table 31: Supplemental rolling 36-month OOS diagnostic.

A.4. Instrument and settlement convention reference

Every candidate option is first translated into a cashflow convention. Equity options record premium, underlying, strike, expiry, right, exercise style, mark, Greek model, payoff source, and split adjustment. VIX options record OSI symbol, VX-forward anchor, VIX settlement source, Black–76 Greek model, and whether expiry P&L used exact VRO/SOQ or a settlement proxy. Short options additionally record assignment-risk flags, simulated margin funding, and short-premium budget use.

A weak implementation-quality score does not automatically exclude a contract. It widens the haircut, raises the stress or capacity penalty, or downgrades the claim. This is why the paper can show VIX option diagnostics without treating proxy settlement as exact listed settlement.

A.5. Proof sketches, solver reductions, and implementation notes

The covariance identity follows from the factor representation. If $r_O = \mu_O + Bf + \varepsilon$, $\text{Cov}(f, \varepsilon) = 0$, $\Omega = \text{Var}(f)$, and $\Sigma_\varepsilon = \text{Var}(\varepsilon)$, then

$$\text{Var}(r_O) = B\Omega B^\top + \Sigma_\varepsilon.$$

The matrix is positive semidefinite because both terms are positive semidefinite. This is the option-risk-model analogue of an equity factor covariance model.

The tangency result is the standard Markowitz argument. Sharpe is homogeneous of degree zero, so impose $q^\top \Sigma_O q = 1$ and maximize $q^\top \mu_O$. The Lagrangian first-order condition gives $\mu_O = 2\lambda \Sigma_O q$, hence $q \propto \Sigma_O^{-1} \mu_O$. Constraints replace this closed-form direction with a fractional program over the feasible set.

The reproducibility package exposes the model code, empirical pipeline, publication tables, figures, strategy-return ledgers, factor regressions, attribution, regime splits, VIX settlement records, claim-audit records, and `tables/empirical_summary.json`. The audit layer independently checks data lineage, point-in-time timing, optimizer constraints, output reproducibility, figure visibility, bibliography scope, caveat text, and LaTeX rendering.

Moment	Value
Median residual mean	0.076
Median residual vol	2.076
Median absolute residual autocorr(1)	0.098
Systematic covariance rank	27.000

Table 32: Training-window residual diagnostics for the local Greek-factor approximation.

The production gap remains visible. Exact VRO/SOQ settlement, executable option quotes, calibrated costs, margin, assignment, borrow, capacity, order routing, and broker reconciliation would be required for production claims. Those claims are not made in this article. The current contribution is the option-only theory object, a point-in-time research implementation, and an audit layer that prevents stronger wording when the data do not support it.